



8th Conference on Blockchain Research & Applications for Innovative Networks and Services

BRAINS 2026

October 13-16, 2026

Florence, Italy

(<https://brains.dnac.org/2026/>)



Workshop on Blockchain & DePIN for Decentralized Energy Systems (*BDePIN-Energy*)

The global energy system is undergoing a profound transformation driven by the proliferation of distributed energy resources (DERs) such as rooftop solar panels, wind turbines, and residential batteries. These *prosumers*—agents who both produce and consume electricity—challenge the traditional unidirectional power grid model, creating an urgent need for decentralized coordination mechanisms that can match local supply and demand efficiently, fairly, and without centralized intermediaries. The April 2025 Iberian Peninsula blackout—triggered by instabilities at large-scale solar farms—underscores the systemic risks inherent in centralized energy architectures and the case for decentralized resilience.

Blockchain technology and Decentralized Physical Infrastructure Networks (DePIN) offer a compelling toolkit to address these challenges. Smart contracts can automate market clearing; AMM-style pricing mechanisms can coordinate prosumers without requiring bilateral negotiations; and tokenized incentive schemes can reward distributed energy contributions at scale. Yet rigorous, interdisciplinary research at the intersection of blockchain infrastructure and energy economics remains nascent. This workshop brings together researchers from distributed systems, economics, mechanism design, and power engineering to advance this frontier.

The BDePIN-Energy workshop invites original research contributions at the intersection of blockchain technology, decentralized physical infrastructure (DePIN), and energy systems.

Topics

Topics of interest include (but are not limited to):

- Automated Market Makers and CFMM-based mechanisms for energy trading
- Smart contract design for peer-to-peer (P2P) energy markets
- DePIN architectures for smart metering, grid telemetry, and sensor verification
- Game-theoretic analysis of prosumer behavior in decentralized energy communities

- Incentive design and tokenomics for distributed energy resources (DERs)
- Mean-field game models of large-scale prosumer populations
- Privacy-preserving protocols for energy data and settlements
- Integration of blockchain with demand-side management and battery optimization
- Empirical studies of existing DePIN and energy blockchain deployments
- Regulatory and governance challenges for decentralized energy markets

Submission guidelines

Papers submitted to *BDePIN-Energy* 2026 Workshop will be assessed based on originality, technical soundness, clarity and interest to a wide audience. All submissions must be written in English and must use standard **IEEE two-column conference template**, available for download from the IEEE website: <https://www.ieee.org/conferences/publishing/templates.html> (<https://www.ieee.org/conferences/publishing/templates.html>)

Workshop papers can be **6–10 pages** (excluding references).

The IEEE template and page-length requirements would not apply to **Presentation-only contributions**, although submissions would still undergo peer review and accepted presenters would be required to register.

Only PDF files will be accepted for the review process and all submissions must be done electronically through EDAS at: <https://edas.info/N35433> (<https://edas.info/N35433>)

Presentation-only submissions are **not subject** to any formatting or page-length requirements. The IEEE BRAINS template is required only for Long, Short, and Demo papers intended for publication in the conference proceedings.

Accepted papers will be published in the BRAINS 2026 proceedings.

Important Deadlines:

- Workshop paper submission deadline: ~~15 July 2026~~ **July 31, 2026 (Extended)**
- Acceptance notification: **1 September 2026**
- Camera-ready paper submission: **15 September 2026**
- Full day workshop: **13 October 2026**

Workshop Chairs



Michele Fabi
(Telecom Paris, CREST, IP Paris, France)



Matheus X.V. Ferreira
(University of Virginia, USA)



Viraj Nadkarni
(Princeton University, USA)

Technical Sponsors



Important dates

- ➔ **Paper Submission Due**
June 14, 2026 (AoE) Firm
- ➔ **Notification of Acceptance**
July 22, 2026 (Extended)
- ➔ **Camera-Ready Papers due**
August 22, 2026 (Extended)

→ **Conference Date**
October 13 - 16, 2026

Submissions



(<https://edas.info/N34965>)

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